Results section for collaborator review

Historical NGC 5548 seasons 1989–1993 14 October 2026

Per season SURD analysis of F5100 and the total, blue, core and red Hβ series with common and uncommon predictor lags, following the September instructions.

1 SURD components in the five historical NGC 5548 seasons

1.1 Series, epochs and analysis design

We apply the seasonal strategy of Lu et al. (2022) and Xi et al. (2025) to the historical monitoring of NGC 5548 between 1988 and 1993. Their seasons are analysed independently; we do the same for the five historical seasons, which carry the labels Year 1 to Year 5 in Table 6 of Lu et al. (2022). The epochs are the intervals of Table 2 of Peterson et al. (2004): JD less 2400000 from 47509 through 47809 (1989), 47861 through 48179 (1990), 48225 through 48534 (1991), 48623 through 48898 (1992) and 48954 through 49255 (1993).

Five observed series are prepared for each epoch: the 5100 Å continuum (F5100), the total broad Hβ flux, and its blue, core and red components. The line measurements are the corrected published columns of Wanders and Peterson (1996), with blue, core and red defined over the observed wavelength intervals 4870–4920, 4920–4960 and 4960–5010 Å. These fixed wavelength intervals play the role of the equal flux velocity bins that Xi et al. (2025) construct over 4700–5050 Å; they are not equal flux bins. Observations on the same date are combined by their mean. The continuum retains its host galaxy contribution. Total and component fluxes come from the same spectra, and the total is the sum of its components, so they are not four independent measurements.

Two variable sets are analysed, each with four members: total, blue, core and red Hβ; and F5100, blue, core and red Hβ. Each member becomes the future target Y in turn. All four members, including the past of the target itself, are the predictors $\mathbf{X}$. The unchanged SURD implementation of Martínez Sánchez et al. (2024) is used throughout. For four predictors the decomposition has four unique components U, eleven redundant components R (six pairs, four triples and one set of four predictors) and eleven synergistic components S with the same combinations. These are 26 components rather than 25. Following Figure 4.1 of Martínez Sánchez (2025), each component is divided by the joint mutual information between the future target and the whole predictor vector,

$$f_a(\tau) = \frac{I_a(Y;\mathbf{X}_\tau)}{I(Y;\mathbf{X}_\tau)}, \qquad \sum_{a=1}^{26} f_a(\tau) = 1, \tag{1}$$

where I_a is the contribution of component a averaged over the states of the target (equations 3.16 of the thesis, ordered as in equation 3.17). The causality leak of equation 3.6 is normalized by the entropy of the target,

$$\ell(\tau) = \frac{H(Y \mid \mathbf{X}_\tau)}{H(Y)}, \tag{2}$$

and shown in its own panel. The component sums equal one to within 1.6×10^{-15} in all 7200 decompositions.

Regular grid and lags. Every series is linearly interpolated onto a daily grid that spans its epoch. Interpolation never extrapolates and never bridges a gap between neighbouring observations longer than 15 days. The target is $Y(t)$ and the predictors are $X_i(t - L - \delta_i)$, with base lag L from 1 through 30 days in one day steps. The 30 day span covers the common lag ranges proposed for the modern seasons. Probabilities are histograms with two bins and, as a sensitivity setting, three bins. Bin boundaries are the marginal quantiles of each full native epoch series. Table 1 lists the data support. Every lag of every curve retains at least 118 tuples. Grid tuples are interpolated and autocorrelated, so they are not independent samples.

Table 1: Data support per epoch. N_F and N_{line} are the numbers of native continuum and spectroscopic dates; the line number is the same for the total, blue, core and red series. The last column gives the range over lags and targets of the daily grid tuples with all five series available (common lags, two bins).

Epoch	JD−2400000	N_F	N_{line}	Grid tuples
1989	47509–47809	124	48	134–196
1990	47861–48179	94	38	161–232
1991	48225–48534	65	32	133–211
1992	48623–48898	83	46	211–244
1993	48954–49255	191	60	231–261

Uncommon predictor lags (equation 3.18). Equation 3.18 of the thesis builds the predictor vector from copies of each variable at different past times. We give every predictor variable its own lag $L + \delta_i$, so that the four predictors still define the 26 components. The common lag analysis has $\delta_i = 0$. In the *blue leading* configuration the blue wing is sampled $\delta = 5$ days further back and in the *red leading* configuration the red wing is. The component that responds earlier to a continuum event carries that event further in its past, so this is the arrangement that aligns the wings under the two kinematic patterns that Xi et al. (2025) describe. The five day offset is a stipulated analysis choice of the size of the uncertainties of published Hβ lags (two to three days); it is not a measured delay. Every configuration and lag is analysed for both variable sets, all four targets and all five epochs: 7200 decompositions, 120 figures.

1.2 Common lags: leakage and dominant components

Figures 1, 2 and 3 show three examples of the 26 normalized components and the leakage against lag. The 120 curves of every epoch, variable set, target and lag configuration are in the three atlas files `atlas_common.pdf`, `atlas_blue_leading.pdf` and `atlas_red_leading.pdf` in `collaborator_rebuild/epoch_results/figures` of the repository, one page per curve. Tables 2 and 3 give the extrema of all 40 common lag curves (two bins).

Table 2: Variable set total, blue, core and red Hβ with common lags and two bins. Leakage minimum and maximum, with the lag in days in parentheses, followed by the largest unique, redundant and synergistic component over lags 1 to 30 d, with its normalized value and lag. H, B, C and R denote the total, blue, core and red series. The maxima are descriptive and have no null calibration or lag uncertainty.

Epoch	Target	$\ell_{\min}$ (d)	$\ell_{\max}$ (d)	Largest U	Largest R	Largest S
1989	Total	0.07 (1)	0.85 (25)	$U(H)$ 0.26 (1)	$R(H,B,C,R)$ 0.64 (7)	$S(B,R)$ 0.45 (27)
1989	Blue	0.16 (1)	0.85 (29)	$U(B)$ 0.36 (1)	$R(H,B,C,R)$ 0.54 (7)	$S(B,C)$ 0.38 (27)
1989	Core	0.07 (1)	0.83 (29)	$U(C)$ 0.27 (1)	$R(H,B,C,R)$ 0.62 (7)	$S(B,R)$ 0.38 (30)
1989	Red	0.17 (1)	0.88 (23)	$U(R)$ 0.22 (1)	$R(H,B,C,R)$ 0.59 (11)	$S(B,C,R)$ 0.28 (23)
1990	Total	0.04 (3)	0.91 (30)	$U(H)$ 0.16 (1)	$R(H,B,C,R)$ 0.66 (1)	$S(B,C,R)$ 0.34 (28)
1990	Blue	0.07 (1)	0.92 (30)	$U(B)$ 0.20 (1)	$R(H,B,C,R)$ 0.67 (5)	$S(B,C,R)$ 0.36 (28)
1990	Core	0.09 (1)	0.92 (28)	$U(C)$ 0.21 (1)	$R(H,B,C,R)$ 0.58 (3)	$S(B,C,R)$ 0.33 (29)
1990	Red	0.09 (1)	0.93 (28)	$U(C)$ 0.17 (15)	$R(H,B,C,R)$ 0.59 (7)	$S(C,R)$ 0.37 (28)
1991	Total	0.06 (1)	0.77 (28)	$U(C)$ 0.25 (28)	$R(H,B,C,R)$ 0.71 (3)	$S(C,R)$ 0.13 (28)
1991	Blue	0.04 (1)	0.74 (29)	$U(C)$ 0.25 (28)	$R(H,B,C,R)$ 0.70 (4)	$S(C,R)$ 0.13 (28)
1991	Core	0.02 (1)	0.68 (30)	$U(C)$ 0.20 (30)	$R(H,B,C,R)$ 0.68 (9)	$S(C,R)$ 0.09 (4)
1991	Red	0.14 (1)	0.83 (24)	$U(R)$ 0.39 (1)	$R(H,B,C,R)$ 0.55 (2)	$S(C,R)$ 0.21 (23)
1992	Total	0.04 (1)	0.94 (30)	$U(R)$ 0.24 (17)	$R(H,B,C,R)$ 0.71 (1)	$S(C,R)$ 0.16 (28)
1992	Blue	0.06 (1)	0.94 (30)	$U(R)$ 0.24 (21)	$R(H,B,C,R)$ 0.76 (2)	$S(B,R)$ 0.19 (28)
1992	Core	0.08 (1)	0.93 (30)	$U(R)$ 0.25 (30)	$R(H,B,C,R)$ 0.69 (2)	$S(H,C,R)$ 0.12 (30)
1992	Red	0.14 (1)	0.96 (30)	$U(R)$ 0.27 (1)	$R(H,B,C,R)$ 0.62 (2)	$S(H,C,R)$ 0.30 (30)
1993	Total	0.23 (1)	0.91 (23)	$U(H)$ 0.46 (1)	$R(H,C)$ 0.40 (5)	$S(H,B,R)$ 0.35 (22)
1993	Blue	0.28 (1)	0.91 (29)	$U(R)$ 0.42 (30)	$R(H,B,C)$ 0.41 (4)	$S(B,C,R)$ 0.44 (25)
1993	Core	0.20 (1)	0.83 (23)	$U(R)$ 0.63 (27)	$R(H,B,C)$ 0.24 (11)	$S(H,B,C,R)$ 0.30 (19)
1993	Red	0.23 (1)	0.81 (13)	$U(R)$ 0.75 (1)	$R(H,B,C,R)$ 0.34 (29)	$S(H,B,C,R)$ 0.25 (16)

Table 3: As Table 2 for the variable set F5100, blue, core and red Hβ. F denotes the continuum.

Epoch	Target	$\ell_{\min}$ (d)	$\ell_{\max}$ (d)	Largest U	Largest R	Largest S
1989	F5100	0.22 (1)	0.71 (18)	$U(F)$ 0.88 (4)	$R(B,C)$ 0.24 (30)	$S(F,B)$ 0.23 (15)
1989	Blue	0.12 (8)	0.45 (29)	$U(F)$ 0.75 (27)	$R(F,B,C,R)$ 0.35 (11)	$S(F,R)$ 0.26 (16)
1989	Core	0.05 (24)	0.40 (30)	$U(F)$ 0.87 (23)	$R(F,B,C,R)$ 0.40 (10)	$S(F,C)$ 0.18 (13)
1989	Red	0.17 (1)	0.50 (30)	$U(F)$ 0.83 (20)	$R(F,B,C,R)$ 0.39 (11)	$S(F,C)$ 0.24 (30)
1990	F5100	0.22 (1)	0.65 (26)	$U(F)$ 0.82 (1)	$R(B,C,R)$ 0.24 (30)	$S(F,B)$ 0.41 (20)
1990	Blue	0.06 (1)	0.63 (30)	$U(F)$ 0.80 (23)	$R(F,B,C,R)$ 0.42 (10)	$S(F,C)$ 0.21 (30)
1990	Core	0.08 (1)	0.71 (30)	$U(F)$ 0.75 (27)	$R(B,C,R)$ 0.44 (1)	$S(F,B)$ 0.25 (12)
1990	Red	0.09 (1)	0.73 (30)	$U(F)$ 0.62 (29)	$R(B,C,R)$ 0.44 (1)	$S(F,C)$ 0.25 (30)
1991	F5100	0.30 (1)	0.85 (18)	$U(F)$ 0.79 (1)	$R(B,R)$ 0.24 (30)	$S(F,B,C,R)$ 0.22 (20)
1991	Blue	0.03 (1)	0.45 (28)	$U(F)$ 0.29 (23)	$R(B,C,R)$ 0.56 (4)	$S(F,B)$ 0.24 (28)
1991	Core	0.00 (4)	0.46 (30)	$U(C)$ 0.22 (1)	$R(B,C,R)$ 0.46 (5)	$S(F,B)$ 0.21 (23)
1991	Red	0.12 (1)	0.64 (26)	$U(F)$ 0.44 (18)	$R(B,C,R)$ 0.41 (1)	$S(F,B,R)$ 0.18 (29)
1992	F5100	0.28 (1)	0.84 (30)	$U(F)$ 0.55 (6)	$R(F,B,C,R)$ 0.29 (1)	$S(F,R)$ 0.34 (30)
1992	Blue	0.06 (1)	0.75 (30)	$U(F)$ 0.62 (21)	$R(F,B,C,R)$ 0.52 (7)	$S(F,B)$ 0.21 (28)
1992	Core	0.07 (1)	0.75 (30)	$U(F)$ 0.41 (17)	$R(F,B,C,R)$ 0.54 (4)	$S(F,C)$ 0.21 (28)
1992	Red	0.11 (1)	0.81 (30)	$U(F)$ 0.61 (16)	$R(F,B,C,R)$ 0.54 (3)	$S(F,B)$ 0.24 (25)
1993	F5100	0.38 (1)	0.84 (30)	$U(F)$ 0.85 (1)	$R(C,R)$ 0.31 (27)	$S(F,B,C,R)$ 0.33 (28)
1993	Blue	0.28 (1)	0.70 (30)	$U(F)$ 0.61 (26)	$R(B,C)$ 0.38 (4)	$S(F,C)$ 0.25 (12)
1993	Core	0.19 (1)	0.61 (19)	$U(C)$ 0.70 (1)	$R(F,R)$ 0.34 (28)	$S(F,R)$ 0.40 (28)
1993	Red	0.24 (1)	0.77 (16)	$U(R)$ 0.87 (1)	$R(B,C,R)$ 0.18 (28)	$S(F,B,C,R)$ 0.36 (13)

Leakage. In 36 of 40 curves the leakage minimum is at the shortest lag of one day, where the history of the target itself predicts its own future. The exceptions are the blue and core targets of the F5100 set in 1989 (minima at 8 and 24 days, leakage 0.12 and 0.05), the total target of the total set in 1990 (3 days) and the core target of the F5100 set in 1991, whose leakage reaches

zero to numerical precision at 4 days. The leakage maximum lies at 20 days or longer in 35 of 40 curves. Median minimum and maximum leakage are 0.12 and 0.70 for the F5100 set and 0.09 and 0.89 for the total set. The leakage at the largest lags of the total set approaches the 0.86 to 0.98 that independent categorical variables produce in the calibration below for 90 to 512 tuples. Little predictive information remains at such lags relative to the bias level of the estimator.

Largest components. At the leakage minimum the largest component is the four-way redundancy $R(H, B, C, R)$ in 16 of the 20 curves of the total set: the components share the same spectra and sum to the total, so their past states carry the same information about the future. In the F5100 set it is a unique component (11 curves, mostly $U(F)$ and the target's own history) or a redundancy among the line components (9 curves). At the leakage maximum the largest component is a synergy in 11 of 20 total set curves and a unique component in five; in the F5100 set it is $U(F)$ in 10 curves and a synergy in seven. The largest component then carries on average 0.32 (total set) and 0.36 (F5100 set) of the joint information, against 0.55 and 0.57 at the leakage minimum.

Continuum to line information. The F5100 set isolates what the continuum adds to a line target beyond the line components: the unique term $U(F)$. For the blue, core and red targets it rises to a maximum of 0.29 to 0.87 (two bins) at lags of 16 to 29 days in 13 of 15 curves (Table 5). The two exceptions have no continuum term above 0.05: the core target in 1991 and the red target in 1993. With three bins the exceptions are the blue and core targets in 1991. A unique continuum component that peaks at a lag of tens of days and leaves leakage near its minimum (0.05 for the 1989 core target at 24 days) is what a delayed response to the continuum would produce. The same peaks give a velocity ordering of the response that we compare with the published seasons below.

Epoch comparison. Mean leakage over lags and targets (common lags, two bins) is 0.49, 0.53, 0.43, 0.58 and 0.65 for 1989 to 1993. These follow the ordering of the grid tuple counts (minimum 134, 161, 133, 211 and 231, respectively), with rank correlation one. Our categorical calibration shows that fewer tuples give a lower estimated leakage even for independent variables, so differences of leakage between epochs mainly reflect sample size.

1.3 Uncommon lags

Table 4 gives the mean leakage of the three lag configurations (two bins, all lags, all four targets of each set) and the change of the allocation. Figure 4 shows an example; the component figures for every configuration are in the atlas files.

Table 4: Mean leakage over the 30 lags and four targets for common lags and for the blue leading and red leading uncommon lags (two bins). The next column is the mean difference blue leading minus red leading over the three line targets. The last column counts line targets for which blue leading has the lower mean leakage.

Epoch	Set	Common	Blue lead	Red lead	$\Delta\ell$	Targets
1989	Total	0.607	0.616	0.605	+0.012	0 of 3
1989	F5100	0.366	0.365	0.355	+0.010	1 of 3
1990	Total	0.613	0.594	0.605	−0.008	2 of 3
1990	F5100	0.448	0.418	0.432	−0.019	3 of 3
1991	Total	0.481	0.458	0.557	−0.098	3 of 3
1991	F5100	0.388	0.398	0.452	−0.058	3 of 3
1992	Total	0.651	0.625	0.695	−0.069	3 of 3
1992	F5100	0.505	0.467	0.515	−0.052	3 of 3
1993	Total	0.702	0.696	0.703	−0.012	1 of 3
1993	F5100	0.601	0.568	0.620	−0.049	3 of 3

Sampling the blue wing five days further back lowers the mean leakage relative to common lags in eight of the ten epoch and set combinations; the red wing does so in four (1989 and 1990), and raises it in 1991 to 1993. The mean differences are small, at most 0.08, but the allocation among

components changes substantially: the median total variation $\frac{1}{2}\sum_a |f_a^{\rm uncommon} - f_a^{\rm common}|$ is 0.30 for both configurations (0.23 to 0.24 for the F5100 set and 0.36 to 0.40 for the total set), and ranges from 0.22 in 1993 to 0.46 in 1989. The lag assigned to each predictor therefore matters for which components receive the information even when the total information changes little. Blue leading gives the lower mean leakage for all three line targets in 1991 and 1992 (both sets) and in the F5100 set in 1990 and 1993; in 1989 it is higher for most targets. This test uses all targets, including the continuum and the target's own history, so it is not a velocity resolved delay measurement.

1.4 Comparison with Lu et al. and Xi et al.

Lu et al. (2022) compile the Hβ lags of the five historical campaigns in their Table 6. Table 5 lists them beside the lag at which the unique continuum component of the blue, core and red targets is largest.

Table 5: Published Hβ lag against continuum ($\tau_{\rm H\beta}$, days; Lu et al. 2022, Table 6) and lag in days of the maximum of $U(F)$, with its normalized value in parentheses, for the blue, core and red targets of the F5100 set with common lags. A dash means $U(F) < 0.05$ at every lag.

		Two bins			Three bins		
Epoch	$\tau_{\rm H\beta}$ (Lu)	Blue	Core	Red	Blue	Core	Red
1989	$19.7^{+1.5}_{-1.5}$	27 (0.75)	23 (0.87)	20 (0.83)	25 (0.49)	27 (0.53)	19 (0.31)
1990	$18.6^{+2.1}_{-2.3}$	23 (0.80)	27 (0.75)	29 (0.62)	19 (0.35)	30 (0.35)	19 (0.35)
1991	$15.9^{+2.9}_{-2.5}$	23 (0.29)	–	18 (0.44)	–	–	13 (0.23)
1992	$11.0^{+1.9}_{-2.0}$	21 (0.62)	17 (0.41)	16 (0.61)	19 (0.26)	30 (0.25)	11 (0.36)
1993	$13.0^{+1.6}_{-1.4}$	26 (0.61)	20 (0.48)	–	27 (0.32)	20 (0.40)	12 (0.15)

The median peak lag over the detected line targets is 23, 27, 20.5, 17 and 23 days (two bins) against the published 19.7, 18.6, 15.9, 11.0 and 13.0 days. It exceeds the published lag in every season by a median of 6.0 days (5.3 days with three bins). The rank correlation between the two sets of lags is 0.67 for two bins and 0.36 for three (five points, not significant). Both show shorter lags in 1991 and 1992 than in 1989 and 1990, and a longer lag in 1993 than in 1992 (17 to 23 days for two bins, 19 to 20 days for three bins). A maximum of a unique information fraction on a one day grid is not the centroid of a cross-correlation function; the information term also depends on the history of the target and on the broad transfer function, and a longer apparent lag is expected when the response is spread over time. The comparison therefore supports a continuum driven response with a delay of the right order in the bright historical state, not an independent lag measurement.

Xi et al. (2025) measure much shorter Hβ lags in the modern low luminosity seasons (Table 6) and show that the velocity resolved lag profile changes between them: an M-shaped profile in 2015, a red-leads-blue pattern in 2019 and 2021 interpreted as inflow, and a blue-leads-red pattern in 2020 and 2023 interpreted as outflow. We read the same ordering from the peak lags of Table 5. With two bins the red wing peaks earlier than the blue wing in 1989 (20 against 27 d), 1991 (18 against 23 d) and 1992 (16 against 21 d), the core peaks earlier than the blue wing in 1993 (20 against 26 d), and the order is reversed in 1990 (blue 23, core 27, red 29 d). With three bins the red peak is again earlier in 1989, 1992 and 1993, and is tied with the blue peak in 1990. A red-leads-blue ordering is therefore found in four of the five historical seasons and survives the change of bins in 1989, 1992 and 1993, similar to the signature that Xi et al. (2025) identify in 2019 and 2021; the season 1990 resembles their blue-leads-red seasons but is not robust to the bin count. The differences between wings are five to seven days, a handful of grid steps, and have no uncertainty. They do not by themselves establish radial inflow: the Wanders and Peterson intervals are fixed wavelength bands and not equal flux bins, the three line series share spectra, and the uncommon lag test of the previous subsection does not single out one wing ordering. The velocity ordering is nevertheless the information theoretic counterpart of the quantity on which Xi et al. (2025) base their kinematic classification, and it varies between seasons in our data as in theirs.

Table 6: Published Hβ lags of the modern seasons and the kinematic classification of their velocity resolved lag profiles (Xi et al. 2025, Table 3 and Section 5.2; seasons before 2023 from Lu et al. 2022, updated by Xi et al. 2025 with a common method; for 2020 Lu et al. 2022 found no credible profile because of the very low variability).

Season	$\tau_{\mathrm{H}\beta}$ (days)	Velocity resolved lag profile
2015	$7.2^{+1.3}_{-0.4}$	wings respond before the core ('M' shape), disk-like
2018	$7.0^{+2.3}_{-3.4}$	no strong ordering
2019	$8.9^{+2.0}_{-1.1}$	red leads blue, inflow-like
2020	$10.0^{+3.3}_{-3.3}$	blue leads red, outflow-like
2021	$9.0^{+1.9}_{-2.5}$	red leads blue, inflow-like
2023	$15.6^{+2.6}_{-2.9}$	blue leads red, outflow-like

The historical lags of 11 to 20 days (Lu et al., 2022) are longer than the 7 to 10 days of 2015 to 2021 and comparable with the 15.6 days of 2023. Xi et al. (2025) show that the broad line region radius follows the optical luminosity with a delay of 3.5 years: the radius fell with the luminosity after the decline that reached its minimum around 2005 and 2010, and approached its typical historical value only by 2023. The short lags of 2015 to 2021 therefore belong to a contracted line region that has not yet recovered, and the historical seasons correspond to the typical radius to which it returned. At the lag of the continuum peak, the leakage of the line targets remains between 0.15 and 0.71 (median 0.54, two bins), so about half of the entropy of a line target is not accounted for by the monitored observables. A driving continuum that differs from the observed 5100 Å flux, as in the disc wind shielding proposed for the 2014 'BLR holiday' (cited in the introduction of Xi et al. 2025), is one possible reason; the noise of the fluxes and the coarse two and three state description are others.

1.5 Sensitivity to the number of bins

Changing the histograms from two to three bins changes the allocation substantially (Table 7). The median total variation $D_{\mathrm{TV}} = \frac{1}{2}\sum_a |f_a^{(2)} - f_a^{(3)}|$ between the bin settings is between 0.27 and 0.58 in all epochs and configurations. For common lags the mean leakage is lower for three bins (0.34 and 0.46 for the F5100 and total sets against 0.46 and 0.61), consistent with the larger finite sample information of sparse histograms. The position of the main peaks can move by ten days or more (Table 5), so the quantities that depend on a single peak lag should be read with this resolution.

Table 7: Median total variation between two and three bin allocations over lags and targets, per epoch, for the F5100 set (first three columns) and the total set (last three), with common, blue leading and red leading lags.

	F5100 set			Total set		
Epoch	Common	Blue lead	Red lead	Common	Blue lead	Red lead
1989	0.37	0.51	0.57	0.39	0.48	0.44
1990	0.32	0.55	0.50	0.37	0.46	0.46
1991	0.55	0.58	0.49	0.49	0.51	0.53
1992	0.27	0.39	0.45	0.37	0.40	0.38
1993	0.42	0.45	0.48	0.36	0.36	0.37

1.6 Limits of the interpretation

The component fractions describe how the available predictive information is distributed; they do not by themselves identify causal coupling or a physical delay. The independent categorical benchmark below shows that finite samples of independent variables produce substantial apparent information, particularly with three states, and that leakage is biased low at small sample sizes. Observation

dates, interpolation across the gaps, flux errors, and the shared spectra of the line series are not included in that benchmark. Lag choices (1–30 d, five day offsets) and the 15 day interpolation limit are fixed in advance and were not tuned. No significance level has been computed for the scans.

Table 8: Median joint information I in bits and normalized leakage ℓ for independent categorical variables (200 draws per setting). The exact values are $I = 0$ and $\ell = 1$. These are not significance thresholds for the AGN data.

Tuples	I, two states	I, three states	ℓ, two states	ℓ, three states
36	0.353	1.304	0.641	0.157
78	0.158	1.076	0.839	0.313
90	0.136	1.015	0.862	0.353
512	0.019	0.268	0.981	0.830

An earlier analysis on the native observation dates, without interpolation, retained only four of 120 curves under a common date requirement across all lags. The regular grid above replaces it so that every variable can be a target in every epoch.

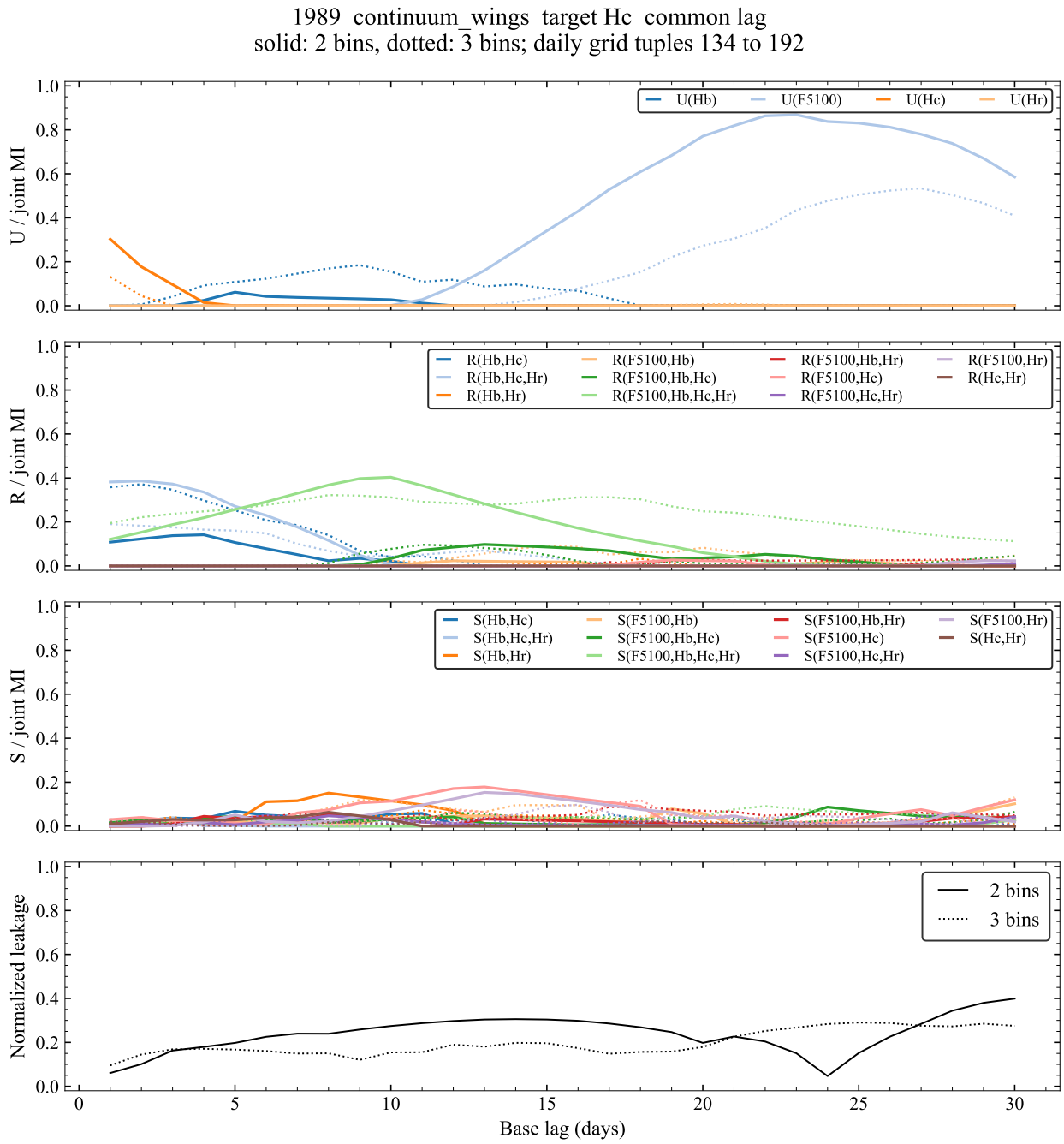


Figure 1: Campaign 1989, variable set F5100, blue, core and red Hβ, future target Core, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 134 to 192 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

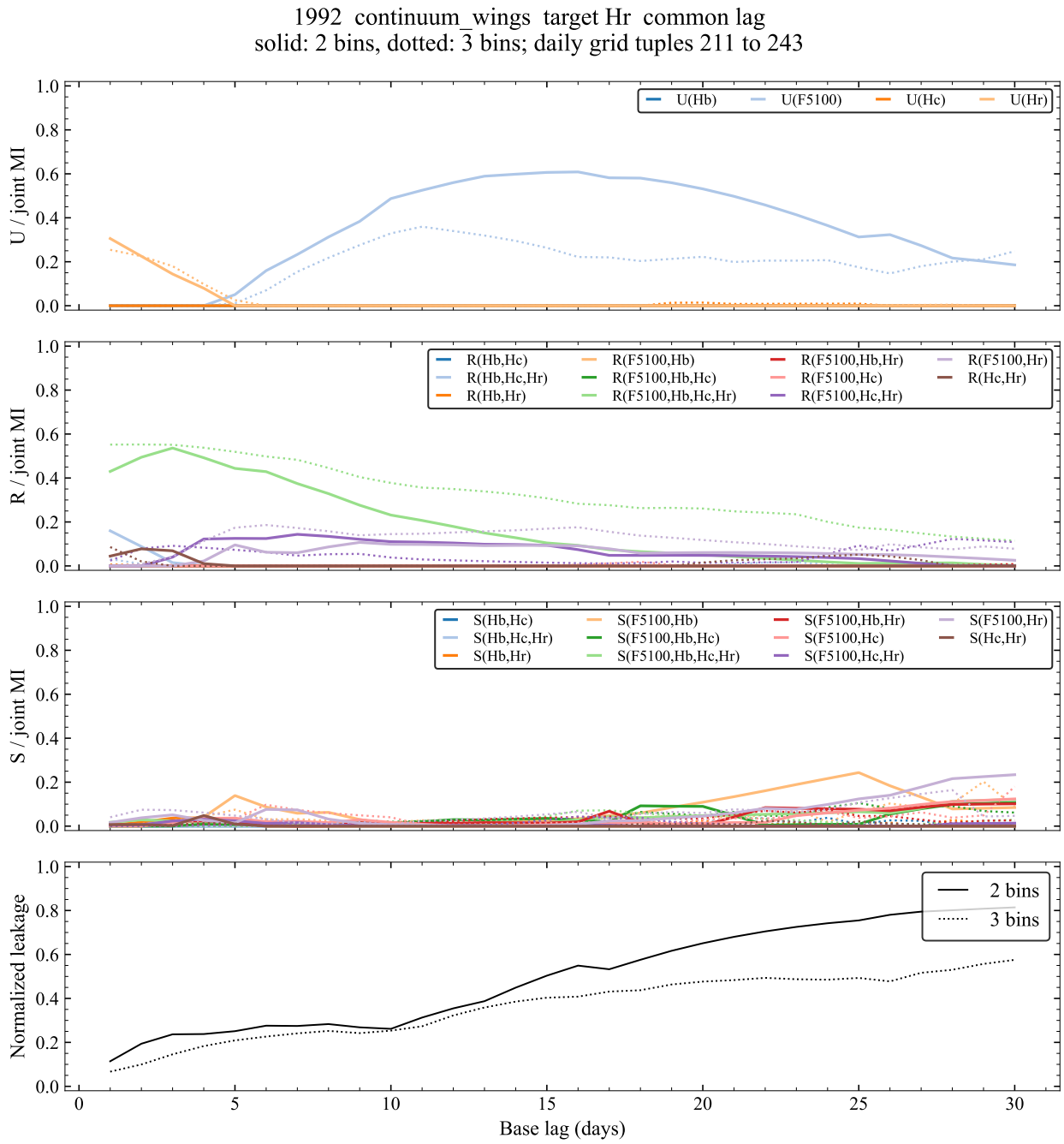


Figure 2: Campaign 1992, variable set F5100, blue, core and red Hβ, future target Red, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 211 to 243 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

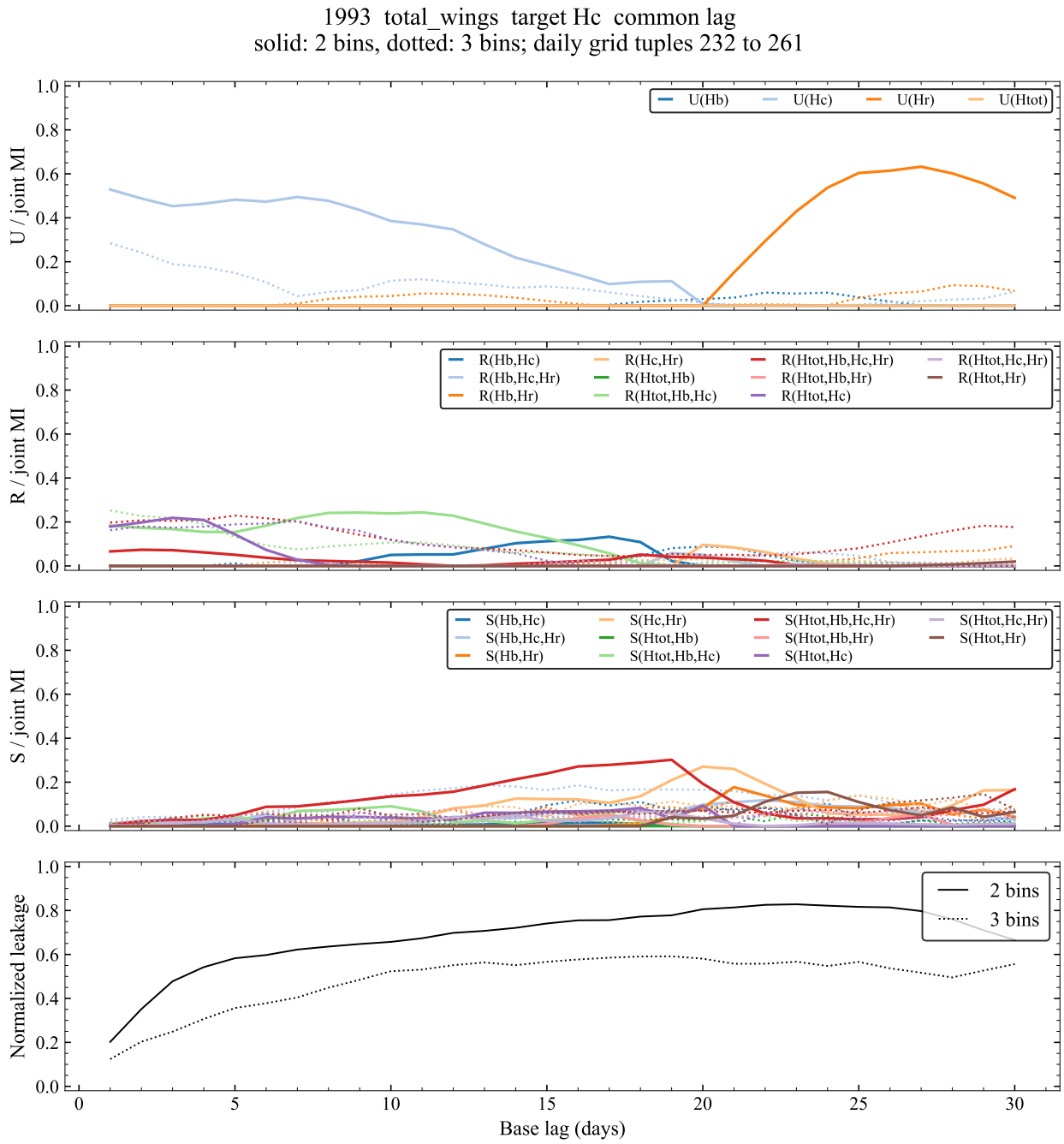


Figure 3: Campaign 1993, variable set total, blue, core and red Hβ, future target Core, common predictor lags. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 232 to 261 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

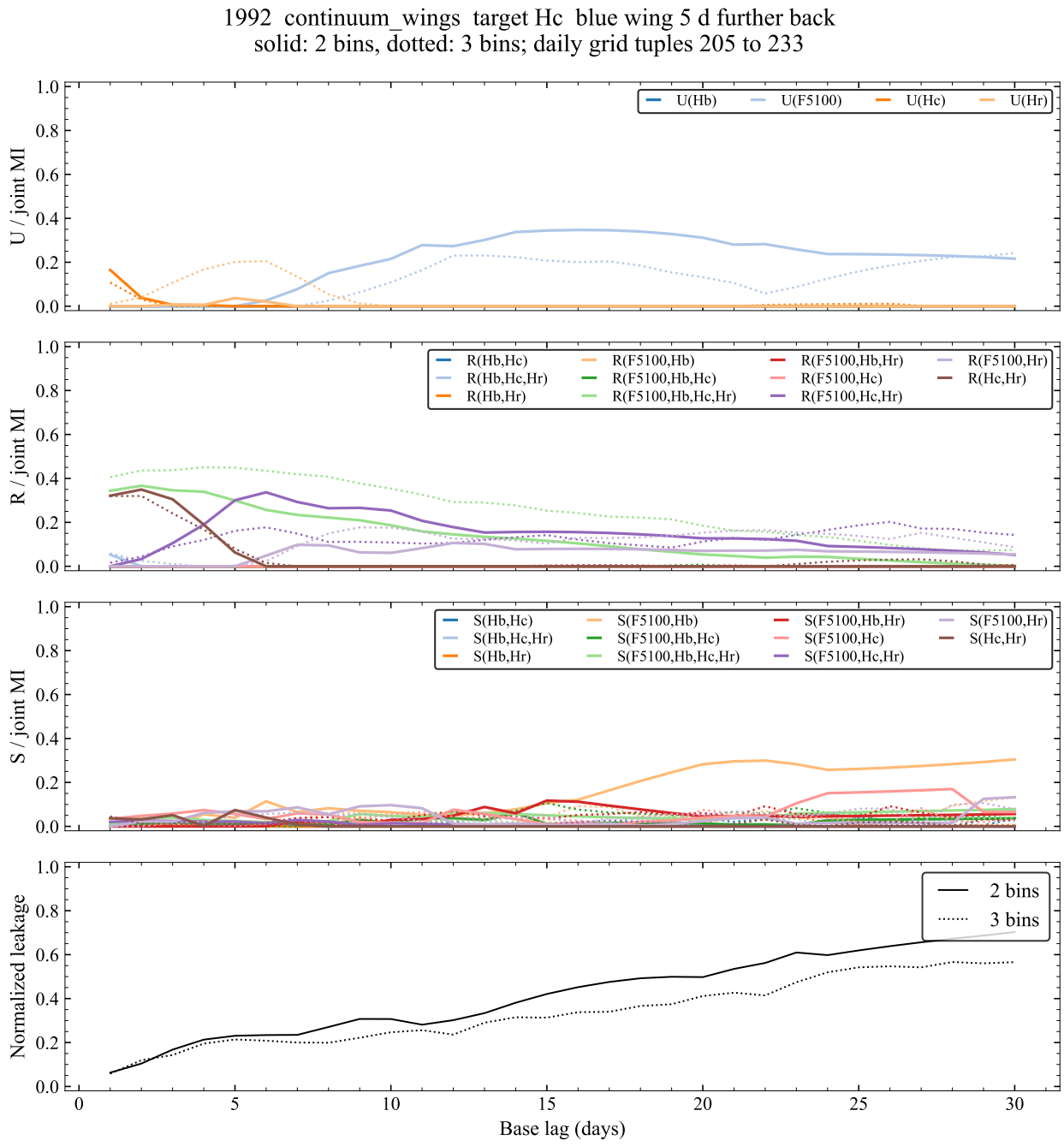


Figure 4: Campaign 1992, variable set F5100, blue, core and red Hβ, future target Core, uncommon predictor lags with the blue wing five days further back. The 26 normalized components (U, R, S panels) and normalized leakage against base lag. Solid curves use two bins and dotted curves three bins; the daily grid supplies 205 to 233 tuples per lag. B, C, R and H or F denote blue, core, red and total or continuum.

References

Wanders, I., and Peterson, B. M. (1996). *A Long Term Study of Broad Emission Line Profile Variability in NGC 5548*. The Astrophysical Journal, 466, 174. doi:10.1086/177502. Corrected measurements are distributed with the AGN Watch archive.

Peterson, B. M., et al. (2004). *Central Masses and Broad Line Region Sizes of Active Galactic Nuclei. II. A Homogeneous Analysis of a Large Reverberation Mapping Database*. The Astrophysical Journal, 613, 682. doi:10.1086/423269.

Martínez Sánchez, Á., Arranz, G., and Lozano Durán, A. (2024). *Decomposing causality into its synergistic, unique, and redundant components*. Nature Communications, 15, 9296. doi:10.1038/s41467-024-53373-4.

Martínez Sánchez, Á. (2025). *Causal inference for complex systems and applications to turbulent flows*. Master of Science thesis, Massachusetts Institute of Technology. Equations 3.6, 3.16, and 3.18 and Figure 4.1.

Lu, K. X., et al. (2022). *Supermassive Black Hole and Broad Line Region in NGC 5548: Results from Five Season Reverberation Mapping*. The Astrophysical Journal Supplement Series, 263, 10. doi:10.3847/1538-4365/ac94d3.

Xi, W. Z., et al. (2025). *Supermassive Black Hole and Broad Line Region in NGC 5548: 2023 Reverberation Mapping Results*. The Astrophysical Journal, 995, 157. doi:10.3847/1538-4357/ae1ccb.